Step 6: corrected QED dependency status

UBT

2026-10-31

AI provenance — Tier B. AI-assisted; machine-verified against the named verifier; individual human attestation is not claimed.

1 Dependency correction

GAP-U1-NEUTRALITY: FORMER PROOF REFUTED; PHYSICAL BRIDGE OPEN. The old left-phase neutrality proof is invalid. Masslessness at a nonzero background is not proved. The scalar comparator instead requires the following condition; it is not a derived UBT kinetic action.

$$m_\gamma^2 = 2e^2\|Qv\|^2, \qquad m_\gamma^2 = 0 \Longleftrightarrow Qv = 0 \quad (e \neq 0).$$

2 Downstream results

Steps 2–5 remain unpaired legacy calculations with unresolved hypotheses, not validated derivations at a charged nonzero UBT background. The standard QED one-loop Schwinger comparison is the following expression. Numerical evaluation of it does not derive a massless photon, Dirac vertex, neutral scalar, Yukawa coupling or vacuum from UBT. The earlier factor-of-two expressions and blanket two-loop estimates in the legacy step 4 are not authoritative. A neutral scalar Yukawa interaction can itself give a one-loop contribution; neutrality does not remove all scalar-loop effects. No universal numerical correction follows without its mass and couplings.

$$a_e^{(1),\mathrm{QED}} = \frac{\alpha}{2\pi}.$$

Neutral-scalar loop reference: `https://arxiv.org/abs/1403.2309`.

3 Verification and open gate

The corrected step 1 and paired research note supply assumptions, exact counterexample, mass normalization, composite response derivation and machine-verification record. The numerical utility `tools/verify_qed_phi_const.py` is a conditional comparison only. It requires an explicit charge and cannot certify UBT closure. LEAN-PENDING: neither lean nor lake is installed. English is the translation source; semantic approval and human attestation are not claimed.